Code No: 132AB **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year II Semester Examinations, January/February - 2024
MATHEMATICS - II
(Common to EEE, ECE, CSE, EIE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) State linearity property of Laplace transform. [2]

b) Find the inverse Laplace transform of $\frac{1}{(s+a)(s+b)}$. [3]

c) If n is a positive integer, then prove that $\Gamma(n+1) = n!$. [2]

d) Prove that $\beta(m,n) = \beta(n,m)$. [3]

e) Evaluate $\int_1^2 \int_1^3 xy^2 dxdy$. [2]

f) Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{dxdy}{1+x^2+y^2}$. [3]

g) Define a direction derivative of a scalar point function in the direction of a vector $\bar{a}$. [2]

h) Define the divergence and curl of a vector point function $\bar{F}$. [3]

i) State Stoke's theorem. [2]

j) If $\bar{F} = (5xy - 6x^2)\bar{i} + (2y - 4x)\bar{j}$, evaluate $\int_C \bar{F}.d\bar{r}$ along the curve C in xy-plane, $y = x^3$ from the point (1, 1) to (2, 8). [3]

PART - B

(50 Marks)

2.a) Find the Laplace transform of the periodic function $f(t) = \begin{cases} \sin \omega t, 0 < t < \frac{\pi}{\omega} \\ 0, \quad \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$.

b) Apply Convolution theorem to evaluate $L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$. [5+5]

OR

3. Solve by transform technique $\frac{d^2y}{dt^2}+2\frac{dy}{dt}-3y=\sin t, y=\frac{dy}{dt}=0$ *when* $t=0$. [10]

4.a) Show that $\beta(m,n+1)+\beta(m+1,n)=\beta(m,n)$.

b) Prove that $\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$. [5+5]

OR

5. Define Beta and Gamma function and hence derive the relation between them. [10]

6. Evaluate $\int_0^{\infty}\int_x^{\infty}\frac{e^{-y}}{y}dxdy$ by changing the order of integration. [10]

OR

7. Change the order of integration in $\int_0^a\int_{x^2}^{2a-x} xy\,dxdy$ and then evaluate it. [10]

8.a) Calculate the angle between the normal's to the surface $xy=z^2$ at the points (4, 1, 2) and (3, 3, -3).

b) Find the value of '*a*' if the vector $(ax^2y+yz)i+(xy^2-xz^2)j+(2xyz-2x^2y^2)k$ has zero divergence. [5+5]

OR

9. If $\bar{r}=x\bar{i}+y\bar{j}+z\bar{k}$, prove that $div(r^n\bar{r})=(n+3)r^n; curl(r^n\bar{r})=0$. [10]

10. Verify Green's theorem for $\int_C\left(x^2+xy\right)dx+\left(x^2+y^2\right)dy$, where C is the square formed by the lines $x=\pm, y=\pm 1$. [10]

OR

11. Verify Stoke's theorem for a vector field defined by $\bar{F}=-y^3\bar{i}++x^3\bar{j}$, in the region $x^2+y^2\le 1, z=0$. [10]

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